Exercise PT-P1: Deep Learning (Neural Networks) (No-Code Text Classification using Neural Networks)

Sayat, Lyona Yvonne A.
Department of Computer Studies and Engineering
Jose Rizal University
Mandaluyong City, Philippines
lyonnayvonne.sayat@my.jru.edu

Abstract*—Machine learning has become one of the most widely used technologies for solving text classification problems because it enables computers to learn from examples rather than relying on manually written rules. In this laboratory exercise, a no-code text classification model was developed using the Machine Learning for Kids platform. The objective was to classify customer feedback into three categories: Positive_Praise, Negative_Complaint, and Urgent_Support. The model was trained using labeled examples and evaluated through a series of testing and stress-testing activities. During the first evaluation, the model achieved an overall accuracy of 80%, correctly predicting eight out of ten unseen test phrases. Error analysis revealed that complaint-related sentences mentioning customer support were incorrectly classified as urgent support requests. To address this limitation, additional representative complaint examples were added to the training dataset before retraining the neural network. After refinement, all test cases were correctly classified, resulting in 100% accuracy with perfect precision, recall, and F1-score for all three classes. This activity demonstrates that the quality, diversity, and representativeness of training data have a direct impact on the performance of supervised machine learning models.

***Keywords*— Machine Learning, Deep Learning, Neural Network, Text Classification, Natural Language Processing, Machine Learning for Kids**

I. Introduction

Machine learning is a branch of artificial intelligence that enables computers to recognize patterns and make predictions using data instead of relying entirely on manually programmed instructions. Unlike traditional software development, where programmers create explicit rules for every possible situation, machine learning allows algorithms to discover patterns automatically from labeled examples. This capability makes machine learning highly effective for natural language processing tasks such as spam filtering, chatbot development, sentiment analysis, recommendation systems, and customer feedback classification.

In this laboratory activity, the Machine Learning for Kids platform was used to build a supervised text classification model without writing any source code. The activity introduced the basic workflow of supervised learning by allowing students to prepare training data, train a neural network model, evaluate its predictions, analyze classification errors, and improve the model through iterative refinement.

The model was designed to classify customer feedback into three categories: Positive_Praise, Negative_Complaint, and Urgent_Support. The exercise emphasized that successful machine learning systems depend not only on the learning algorithm itself but also on the quality and diversity of the dataset used for training.

II. Methodology

The Machine Learning for Kids platform was used to develop a supervised text classification model without writing any source code. Three classification labels were created: Positive_Praise, Negative_Complaint, and Urgent_Support.

Initially, ten training examples were prepared for each category. Positive_Praise contained customer compliments and positive shopping experiences. Negative_Complaint included examples related to damaged products, delayed deliveries, refund requests, poor product quality, and customer service complaints. Urgent_Support consisted of account access problems, verification issues, password reset requests, payment errors, and technical support requests requiring immediate assistance.

After preparing the dataset, the platform automatically trained a neural network model. During training, the system analyzed the labeled examples, extracted language features, and learned relationships between words and their corresponding categories. Once the training process was completed, the model was evaluated using ten unseen test phrases to measure its classification performance. Finally, the model was refined by adding additional complaint-related examples and retraining the neural network to improve prediction accuracy.

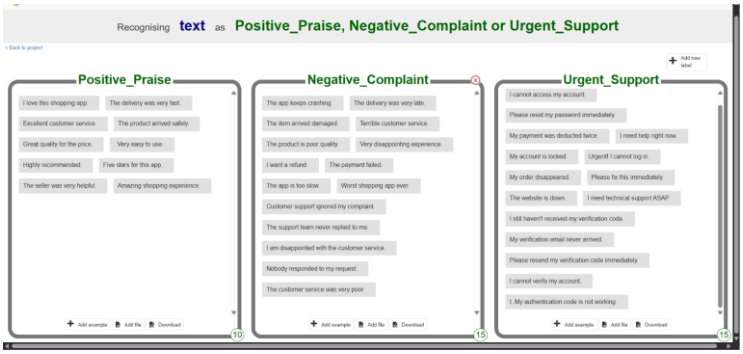


Fig. 1. Initial training dataset containing Positive_Praise, Negative_Complaint, and Urgent_Support examples.

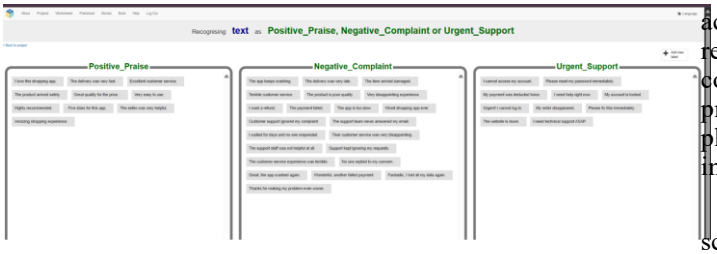


Fig. 2. Refined training dataset after adding additional complaint-related examples to improve model performance.

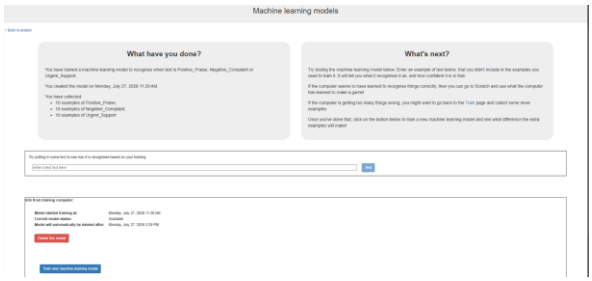


Fig. 3. Trained machine learning model showing the training date, model status, and the number of training examples for each classification category.

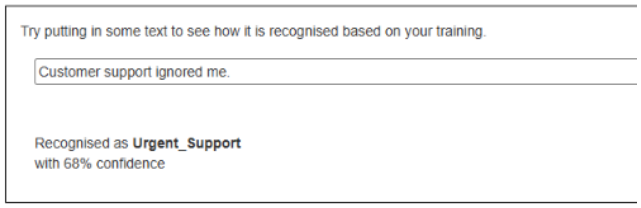


Fig. 4. Initial prediction before model refinement showing the incorrect classification of "Customer support ignored me." as Urgent_Support with 68% confidence.

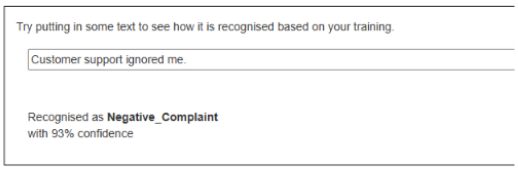


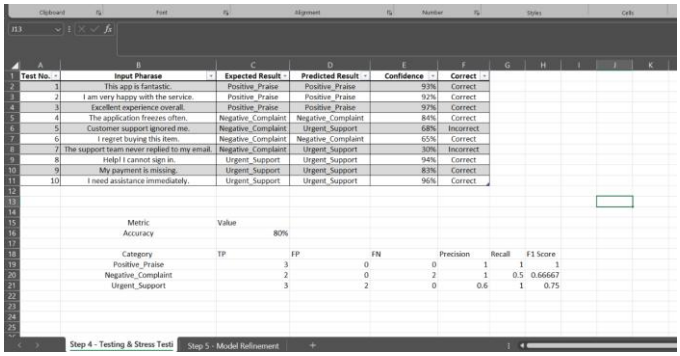
Fig. 5. Prediction after model refinement showing the correct classification of "Customer support ignored me." as Negative_Complaint with 93% confidence.

III. MODEL TESTING AND EVALUATION

The trained model was evaluated using ten test phrases that were intentionally excluded from the training dataset. These phrases consisted of positive feedback, complaint statements, and urgent support requests. The purpose of the evaluation was to determine whether the model had learned general language patterns instead of memorizing the training examples.

During the initial testing phase, eight of the ten test phrases were correctly classified, resulting in an overall accuracy of **80%**. Positive feedback and urgent support requests were recognized with high confidence scores, while complaint-related sentences mentioning customer support produced incorrect predictions. The model associated the phrase "customer support" with urgent technical assistance instead of customer dissatisfaction.

Performance evaluation using Precision, Recall, and F1-score further demonstrated that the model performed well overall but required additional complaint examples to reduce confusion between Negative_Complaint and Urgent_Support



Test No.	Input Phrase	Expected Result	Predicted Result	Confidence	Correct
1	This app is fantastic.	Positive_Praise	Positive_Praise	93%	Correct
2	I am very happy with the service.	Positive_Praise	Positive_Praise	92%	Correct
3	Excellent experience overall.	Positive_Praise	Positive_Praise	97%	Correct
4	The application freezes often.	Negative_Complaint	Negative_Complaint	84%	Correct
5	Customer support ignored me.	Negative_Complaint	Urgent_Support	68%	Incorrect
6	I regret buying this item.	Negative_Complaint	Negative_Complaint	65%	Correct
7	The support team never replied to my email.	Negative_Complaint	Urgent_Support	30%	Incorrect
8	Help! I cannot sign in.	Urgent_Support	Urgent_Support	94%	Correct
9	My payment is missing.	Urgent_Support	Urgent_Support	83%	Correct
10	I need assistance immediately.	Urgent_Support	Urgent_Support	96%	Correct

Metric	Value
Accuracy	80%

Category	TP	FP	FN	Precision	Recall	F1 Score
Positive_Praise	3	0	0	1	1	1
Negative_Complaint	2	0	2	1	0.5	0.66667
Urgent_Support	3	2	0	0.6	1	0.75

Table I. Initial Model Performance

IV. MODEL REFINEMENT

To improve the model, additional training examples focusing on customer support complaints and sarcastic complaint expressions were added to the Negative_Complaint category. These examples exposed the neural network to a wider variety of complaint-related language patterns.

After retraining the model, the previously misclassified sentences were evaluated again. Both "Customer support ignored me." and "The support team never replied to my email." were correctly classified as Negative_Complaint. Their confidence scores also increased significantly, demonstrating that the additional training examples improved the model's ability to distinguish complaints from urgent support requests.

Following refinement, the model correctly classified all ten test phrases, increasing the overall accuracy from **80%** to **100%**. Precision, Recall, and F1-score also reached perfect values for all three categories, indicating a significant improvement in classification performance.



Test No.	Input Phrase	Expected Result	Predicted Result	Confidence	Correct
1	This app is fantastic.	Positive_Praise	Positive_Praise	93%	Correct
2	I am very happy with the service.	Positive_Praise	Positive_Praise	92%	Correct
3	Excellent experience overall.	Positive_Praise	Positive_Praise	97%	Correct
4	The application freezes often.	Negative_Complaint	Negative_Complaint	84%	Correct
5	Customer support ignored me.	Negative_Complaint	Negative_Complaint	93%	Correct
6	I regret buying this item.	Negative_Complaint	Negative_Complaint	65%	Correct
7	The support team never replied to my email.	Negative_Complaint	Negative_Complaint	96%	Correct
8	Help! I cannot sign in.	Urgent_Support	Urgent_Support	94%	Correct
9	My payment is missing.	Urgent_Support	Urgent_Support	83%	Correct
10	I need assistance immediately.	Urgent_Support	Urgent_Support	96%	Correct

Metric	Value
Accuracy	100%

Category	TP	FP	FN	Precision	Recall	F1 Score
Positive_Praise	3	0	0	1	1	1
Negative_Complaint	4	0	0	1	1	1
Urgent_Support	3	0	0	1	1	1

Table II. Refined Model Performance

V. DISCUSSION

A. Explicit Rules vs. Tagged Learning

If this application had been implemented using explicit IF/ELSE statements, every possible customer sentence would require a manually written rule. Because customers express the same concern using different words, grammar, abbreviations, and writing styles, the number of rules would quickly become extremely large and difficult to maintain. New expressions would often be classified incorrectly unless programmers continuously updated the code.

Machine learning solves this limitation by allowing the neural network to learn directly from labeled examples. Instead of memorizing exact rules, the model recognizes language patterns that allow it to classify previously unseen sentences. This activity demonstrated that tagged learning is more flexible, scalable, and suitable for text classification than traditional rule-based programming.

B. Data Bias and Dataset Quality

The experiment clearly showed that dataset quality has a direct impact on model performance. During the first evaluation, the model achieved only 80% accuracy because complaint-related examples involving customer support were underrepresented in the training dataset. As a result, the model incorrectly associated customer support phrases with urgent technical assistance.

Although this experiment was ultimately trained using ten examples per class, using only three examples per class would likely result in lower accuracy because the neural network would have fewer language patterns to learn from. With a limited number of examples, the model would struggle to recognize different ways customers express similar concerns, making it more likely to confuse related categories such as Negative_Complaint and Urgent_Support. Increasing the training data to ten examples per class and including more representative and diverse examples enabled the model to learn more reliable language patterns and improve its ability to classify previously unseen sentences.

After additional complaint examples were added and the model was retrained, both previously incorrect predictions became correct, increasing the overall accuracy from 80% to 100%. This demonstrates that deep learning models require balanced, representative, and diverse datasets to achieve reliable performance. Increasing both the quantity and quality of training examples allows the model to generalize better when processing unseen data.

C. Connecting to the Slide Architecture

Although Machine Learning for Kids does not display the internal architecture of the neural network, feature extraction most likely occurred immediately after the text input was processed and before the final classification stage. During this step, the platform converted the input text into numerical representations that the neural network could analyze.

Instead of processing complete sentences as plain text, the model learned meaningful language patterns such as important keywords, word combinations, and contextual relationships. Words such as *excellent*, *refund*, *crashing*, *password*, and *immediately* became useful features that helped distinguish between **Positive_Praise**, **Negative_Complaint**, and **Urgent_Support**. The improvement in prediction accuracy after retraining indicates that adding more representative training examples enabled the feature extraction process to learn stronger patterns, resulting in more accurate classifications.

VI. CONCLUSION

The laboratory exercise successfully demonstrated the complete workflow of developing a supervised text classification model without writing program code. Using the Machine Learning for Kids platform, a neural network was trained to classify customer feedback into Positive_Praise, Negative_Complaint, and Urgent_Support. Initial testing produced an accuracy of 80%, revealing weaknesses in classifying complaint-related customer support messages. By refining the training dataset and adding more representative complaint examples, the model achieved perfect classification performance with 100% accuracy.

The activity emphasized that the success of machine learning models depends not only on the algorithm but also on the quality, diversity, and representativeness of the training data. It also provided practical experience in supervised learning, model evaluation, feature learning, and iterative model improvement. Overall, the exercise strengthened the understanding of deep learning concepts and demonstrated how no-code platforms can simplify the development of intelligent text classification systems.

[1] Machine Learning for Kids, “Machine Learning for Kids.” Available: https://machinelearningforkids.co.uk/. Accessed: Jul. 29, 2026.

[2] D. Jurafsky and J. H. Martin, Speech and Language Processing, 3rd ed., Pearson, 2024.